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Contractor Report
CR-NAVFAC EXWC-CIOFP-2129

2021 WATER TANK INSPECTION AND ASSESSMENT AT NAVAL WEAPONS STATION EARLE COLTS NECK, NEW JERSEY

By
TANK INDUSTRY CONSULTANTS
June 2021

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14. ABSTRACT
This report presents health, safety and structural findings, repair recommendations, and cost estimates for Tank 558 located at the Naval Weapons Station Earle Colts Neck, New Jersey

15. SUBJECT TERMS
Observations, recommendations, and cost estimates relating to safety, sanitary, operational, and structural items as well as coatings and seismic upgrades (such as piping flexibility and freeboard).

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2021 WATER TANK INSPECTIONS AND ASSESSMENTS AT

NAVAL WEAPONS STATION EARLE COLTS NECK, NEW JERSEY

CR-NAVFAC EXWC-CIOFP-2129

JUNE 2021

by:
Tank Industry Consultants
7740 West New York Street
Indianapolis, Indiana 46214

under:
Contract No. N39430-16-D-1815

Prepared for:
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WATER TANK INSPECTIONS AND ASSESSMENTS NAVAL WEAPONS STATION EARLE COLTS NECK, NEW JERSEY

EXECUTIVE SUMMARY

This report was prepared as part of the Water Tank Inspection Program administered by the Naval Facilities Engineering and Expeditionary Warfare Center (NAVFAC EXWC).

This program provides safety, sanitary, and structural inspection for water tank facilities. All of these services, including this report, were provided by Tank Industry Consultants (TIC), under the responsible charge of Gregory R. "Chip" Stein, P.E., in accordance with Contract No. N39430-16-D-1815.

One tank was evaluated by Tank Industry Consultants in June 2021. Tank 558 is a ground-level welded steel tank located at the Naval Weapons Station Earle in Colts Neck, New Jersey.

The purpose of this report is to present the findings of the comprehensive inspections and to make recommendations for the repair, recoating, structural condition, and maintenance of the tanks, including budgetary estimates to perform repair work. The following is a summary of inspection findings and identified deficiencies that should be rectified for full compliance with industry standards. Organizations and standards cited throughout the report are: American National Standards Institute (ANSI), American Petroleum Institute (API), American Water Works Association (AWWA), American Concrete Institute (ACI), American Society of Civil Engineers (ASCE), Occupational Safety and Health Administration (OSHA), and The Society for Protective Coatings (SSPC).

<u>Tank 558</u>

Tank 558 is a ground-level welded steel tank storing 300,000 gallons of non-potable water used exclusively for the purposes of fire protection for the Naval Weapons Station Earle in Colts Neck, New Jersey.

This tank received a structural condition rating of 83.

The coating on the exterior of the tank appears to be in fair to poor overall condition as corrosion is located on several areas of the roof and shell. Tank Industry Consultants recommends the exterior should be recoated within the next 2 to 3 years. The interior surfaces should be drained and inspected before the rehabilitation of the tank. The interior coating appears to be in fair to poor overall condition as corrosion is located on areas of the roof, support structure, and shell. Tank Industry Consultants recommends the interior should be recoated within the next 2 to 3 years.

OSHA and Safety-Related Deficiencies:

- the valve vault was not locked prior to or after the field evaluation,
- the electrical outlets at the water level indicating device equipment on the roof are not equipped with ground fault interrupt circuits,
- the size of the shell manhole is too small (AWWA D100-11 Sec. 7.4.4),
- the tank is equipped with only one shell manhole (AWWA D100-11 Sec. 7.4.4),
- the exterior shell ladder is not equipped with a vandal deterrent,
- the roof safety railing access opening is not equipped with a self-closing gate (29 CFR 1910.28(b)(3)(iv)), and
- the gap between the roof safety railing toe bar and the roof is too large (29 CFR 1910.29(k)(1)(iii)).

It is recommended that these deficiencies be rectified to be in compliance with OSHA and safety-related standards.

Operational and AWWA Deficiencies:

- the overflow pipe discharges near the top of the shell which allows the potential of ice-formation of the effluent on the shell during freezing weather,
- the projection of the stub overflow pipe from the shell is too small (AWWA D100-11 Sec. 7.3),
- the roof manhole was not locked prior to or after the field evaluation (AWWA D100-11 Sec. 7.4.3), and
- the roof vent is not clog-resistant (AWWA D100-11 Sec. 7.5).

These deficiencies should be corrected.

If the work is not performed within the next 2 to 3 years, the structure should be reevaluated prior to the preparation of specifications and the solicitation of bids. Proper maintenance after completing the recommendations listed in this report would include periodic washouts and evaluations approximately every 3 to 5 years.

WATER TANK
INSPECTIONS AND ASSESSMENTS AT

NAVAL WEAPONS STATION EARLE COLTS NECK, NEW JERSEY

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1. INTRODUCTION

1.1 BACKGROUND/OBJECTIVES

This report was prepared as part of the Water Tank Inspection Program administered by the Naval Facilities Engineering and Expeditionary Warfare Center, NAVFAC EXWC.

This program provides inspection, assessment, repair recommendations, and estimates of repair costs for water tank facilities. All of these services, including this report, were provided by Tank Industry Consultants, under the responsible charge of Gregory R. “Chip” Stein, P.E., in accordance with Contract No. N39430-16-D-1815.

One tank was evaluated by Tank Industry Consultants on June 21, 2021 under the responsible charge of Chad D. Mitchell, NACE-Certified Coating Inspector, Level 2 (#56876). Additional key personnel involved with the evaluation and report are listed in Appendix A.

Tank 558 is a ground-level welded steel tank located at the Naval Weapons Station Earle in Colts Neck, New Jersey.

INSPECTION BRIEFINGS

The following individuals were present at the inspection briefing and/or the field evaluations:

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2. ACTIVITY DESCRIPTION

TABLE 2-1
TANKS INSPECTED

Tank No.	Capacity (Gallons)	Type	Dimensions	Water Purpose	NFA
558	300,000	Ground-Level Welded Steel	Approx. 50 ft (15.2 m) diameter, Approx. 22 ft (6.7 m) shell height	Non-Potable Fire Protection	NFA100000927973

3. TANKS INSPECTED

The following sections contain discussion of observations from the field evaluations and recommendations for rehabilitation of the tanks. Each section is divided into the following sub-sections

1. Description of the Facilities
2. Observed Conditions
3. Comparison of Previous Inspection Results
4. Structural Condition Assessment
5. Recommendations

The inspection procedure is presented in Appendix B and structural data appears in Appendix C.

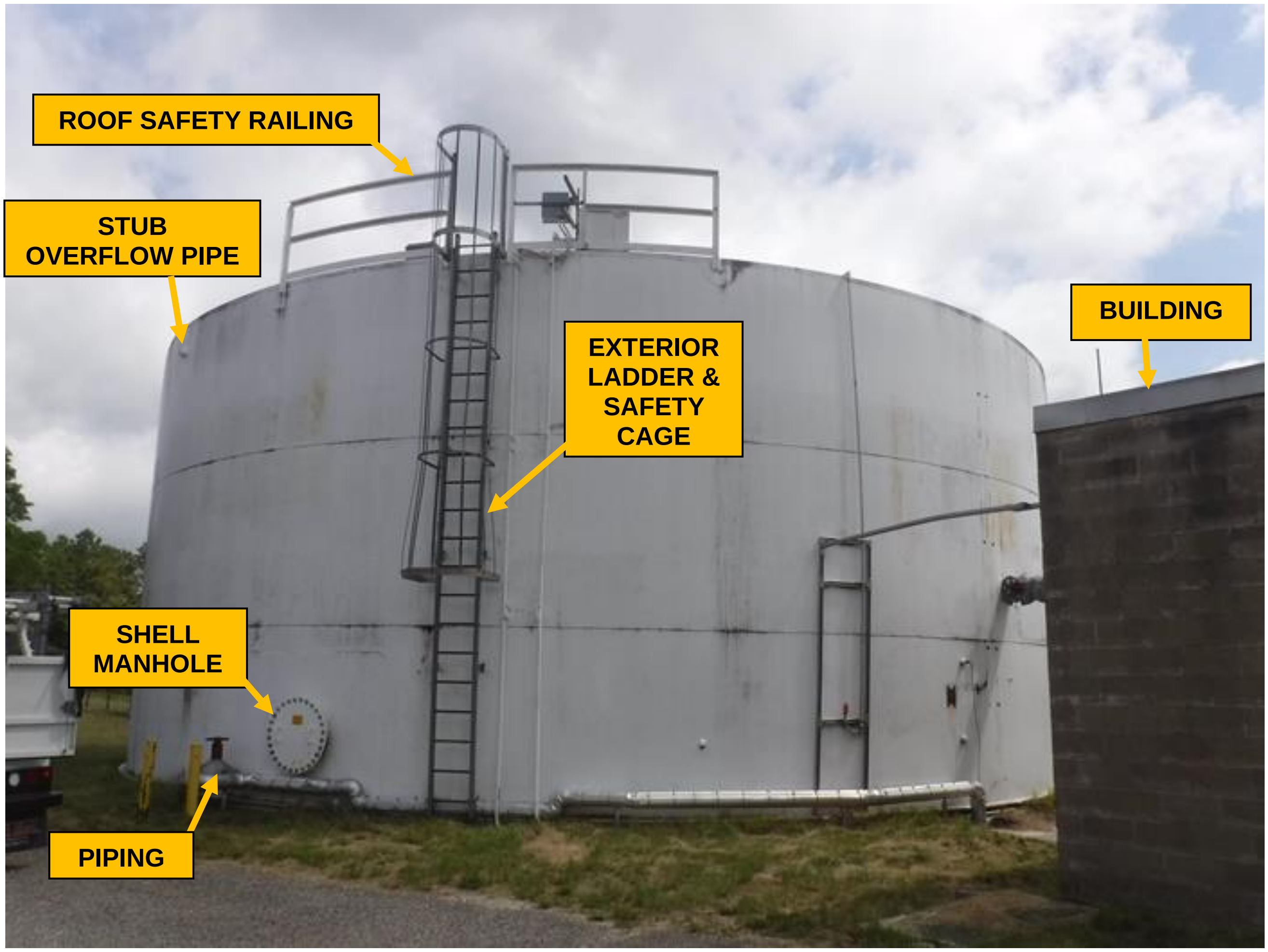
3.1 TANK 558

Tank 558 is located at the Naval Weapons Station Earle in Colts Neck, New Jersey, and stores non-potable water used exclusively for the purposes of fire protection.

3.1.1 Description of the Facility

Tank 558 is a 300,000 gallon welded steel ground storage tank located at the Naval Weapons Station Earle in Colts Neck, New Jersey. According to information on the tank information stencil, the tank has a capacity of 300,000 gallons. Measurements taken during the field evaluation indicate the tank is approximately 50 ft (15.2 m) in diameter and has a shell height of approximately 22 ft (6.7 m). (See Figure 3.1-1)

TANK 558



Distribution limited to the DoD and U.S. DoD contractors only; administrative/operational use June 2021.	TIC TANK INDUSTRY CONSULTANTS Contract No. N39430-16-D-1815	**NAVAL FACILITIES ENGINEERING COMMAND** **ENGINEERING AND EXPEDITIONARY WARFARE CENTER** **Fig. No. 3.1-1**

3.1.2 Observed Conditions

Tank 558 is located at the Naval Weapons Station Earle in Colts Neck, New Jersey (See photos 3.1-1 to 3.1-4). A fenced storage yard and open areas are adjacent to the tank site. The nearest overhead power lines are located west of the tank on the site.

The tank site is covered with well-maintained grass and is flat. The immediate tank site is not fenced. A building and antenna tower enclosed within a chain link fence topped with barbed wire are located west of the tank on the site (See photo 3.1-4). Building 559 is located south of the tank on the site (See photo 3.1-3).

A valve vault is located south of the tank on the site (See photos 3.1-5 to 3.1-7). Areas of the valve vault extend below the tank and Building 559. The access into the valve vault was not locked prior to or after the field evaluation. Several pipes inside the valve vault are covered with jacketed insulation which restricted their evaluation. Widespread corrosion is present on the uncovered piping inside the valve vault (See photos 3.1-6 and 3.1-7). The vault is not equipped with a sump or pump, and water is present on the vault floor. The valve vault was not locked prior to or after the field evaluation which is a safety-related deficiency.

The tank rests on a concrete foundation (See photo 3.1-8). The foundation does not meet the AWWA recommended minimum projection of 6 in. (152 mm) and is mostly buried. The visible areas of the foundation appear to be in adequate condition. A white colored coating is visible on the foundation. **No grout or sealant is present between the foundation and bottom plate, and gaps are located between the foundation and bottom plate (See photo 3.1-9).**

The floor of the tank may have been previously replaced as the tank is equipped with two bottom plates (See photos 3.1-8 to 3.1-10). The upper bottom plate is located approximately 4 in. (102 mm) above the lower bottom plate. Corrosion is present on random areas along the upper bottom plate and minor metal loss is present along the lower bottom plate. A flat bar stiffener is located just below the upper bottom plate and is seam welded along the top side only. Grass clippings and leaves are located on the bottom plates.

The shell’s contour appears to be in adequate condition. The coating on the exterior shell is in fair overall condition as corrosion is present on areas of the shell (See photos 3.1-20 and 3.1-22). The coating has chalked, and drips, sags, and debris are present in the coating. Mildew is located on areas of the shell. The exterior shell coating has fair adhesion to the underlying coating. Several unused stud remains are located on the south side of the shell, and coating failure, corrosion, and rust staining are present on the areas where cabinets were previously located (See photos 3.1-20 to 3.1-22). An approximately 5 ft 6 in. (1.7 m) x 9 ft 1 in. (2.7 m) welded steel insert plate with rounded corners is located in the shell around the shell manhole and pipe penetrating near the bottom of the shell (See photo 3.1-18). Two threaded and plugged couplings are located on the south side of the shell (See photo 3.1-19). Another threaded and plugged coupling is located on the south side of the shell, and a wire extends from the valve vault and connects to this coupling (See photos 3.1-13 and 3.1-14). Two electrical conduits extend along the shell between the two bottom plates on the south side. Three electrical conduits extend up the shell on the southwest side adjacent to the exterior shell ladder, and corrosion is present on the conduit brackets (See photos 3.1-25 and 3.1-26). Another conduit extends up the shell on the south side (See photos 3.1-23 and 3.1-24).

Capped water level indicating device equipment penetrates near the edge of the roof and is located within an uncovered enclosure (See photo 3.1-34). The device is equipped with electrical outlets which do not include ground fault interrupt circuits. Four capped pipes project from the shell between the two bottom plates (See photo 3.1-10). A pipe equipped with a valve and covered with jacketed insulation extends from the valve vault, along the site, and penetrates near the bottom of the shell on the west side (See photos 3.1-13 and 3.1-15). Two pipe bollards are adjacent to this pipe penetration. A pipe equipped with a valve and covered with jacketed insulation extends from Building 559 and penetrates the tank shell (See photos 3.1-11 and 3.1-12). The shell plate surrounding the pipe penetration is equipped with a reinforcing plate, and an unplugged weep hole is located in the reinforcing plate (See photo 3.1-12). The electrical outlets at the water level indicating device equipment on the roof are not equipped with ground fault interrupt circuits which is a safety deficiency.

The tank is equipped with one flanged and bolted circular shell manhole located on the west side of the tank (See photo 3.1-16). A tank information decal and two handles are located on the cover of the shell manhole (See photo 3.1-17). The cover for the flanged and bolted manhole is not equipped with a hinged support on the tank exterior. The shell plate surrounding the manhole is not equipped with a reinforcing plate. Spots of coating failure, corrosion, and rust staining are located around the shell manhole bolts. The shell manhole bolts and nuts are not equipped with washers. Safety-related and AWWA deficiencies include: (1) the 24 in. (610 mm) diameter manhole does not meet the minimum required 30 in. (762 mm) diameter size (AWWA D100-11 Sec. 7.4.4), and (2) the tank is equipped with only one shell manhole (AWWA D100-11 Sec. 7.4.4).

The stub overflow pipe exits near the top of the shell and immediately discharges (See photo 3.1-27). The discharge end of the overflow pipe is equipped with a perforated plate screening. No significant coating failure or corrosion are present on the overflow pipe. AWWA and operational deficiencies include: (1) the overflow pipe discharges near the top of the shell which allows the potential of ice-formation of the effluent on the shell during freezing weather, and (2) the 3 in. (76 mm) projection of the overflow pipe from the shell does not meet the minimum required 12 in. (305 mm) projection (AWWA D100-11 Sec. 7.3).

An exterior ladder provides access from the ground to the tank roof (See photos 3.1-28 to 3.1-32). The ladder is not equipped with a safe-climbing device; however, OSHA does not require safe-climbing devices on ladders less than 24 ft (610 mm) high. The ladder is equipped with a safety cage constructed of welded and bolted flat bar members. The base of each safety cage section is flared. The ladder is bolted to brackets which are welded to the shell. Widespread corrosion is present on the ladder rungs, rust staining is present on the ladder side rails, and corrosion is present on the brackets (See photos 3.1-29 and 3.1-30). The exterior shell ladder is not equipped with a vandal deterrent which is a safety-related deficiency.

A roof safety railing is located at the shell ladder access to the roof and adjacent to the roof manhole (See photos 3.1-32 and 3.1-33). The roof safety railing is constructed of welded angle and flat bar members. No significant coating failure or corrosion is present on the safety railing. An electrical cabinet is attached to the roof safety railing, and the mid-rail is replaced with a bracket in this area (See photo 3.1-33).

Wiring extends from the cabinet and penetrates a hand hole cover plate on the roof (See photo 3.1-35). Safety-related and OSHA deficiencies include: (1) the roof safety railing access opening is not equipped with a self-closing gate (29 CFR 1910.28(b)(3)(iv)), and (2) the 1/2 in. (13 mm) gap between the roof safety railing toe bar and the roof is more than the maximum allowed 1/4 in. (6 mm) (29 CFR 1910.29(k)(1)(iii)).

The contour of the roof is adequate. The exterior roof coating appears to be in fair to poor condition as corrosion is located on several areas of the roof (See photos 3.1-41 to 3.1-43). The roof coating has chalked, and debris is present in the coating. The roof coating has good adhesion to the underlying coating. Twenty circular hand holes and cover plates are located on the roof (See photos 3.1-38 to 3.1-40). A conduit extends across the roof, and wiring from the conduit junction boxes penetrates hand hole cover plates on the roof (See photos 3.1-38 to 3.1-40). Several unused conduit brackets are bolted to the roof. Seventeen rectangular cover plates are bolted to the roof, and corrosion is present on the plates (See photo 3.1-46). These plates are equipped with gaskets, but the gaskets have deteriorated. Several bolts are missing from these plates. Threaded and plugged couplings are located on the roof, and corrosion is present on the couplings (See photo 3.1-47). Nine studs are located in the roof, and sealant is located around the bolts (See photos 3.1-44 and 3.1-45).

The roof is equipped with one hinged-cover manhole which is welded on the exterior and interior (See photos 3.1-36 and 3.1-37). Significant corrosion is present on the interior of the roof manhole cover and along the curb. Wires penetrate the roof manhole curb. The roof manhole was not locked prior to or after the field evaluation which is a safety-related and AWWA deficiency (AWWA D100-11 Sec. 7.4.3).

A mushroom-cover vent is located near the center of the roof (See photos 3.1-48 to 3.1-50). The roof vent neck is equipped with a flat bar stiffener (See photo 3.1-49). The roof vent is equipped with horizontally-oriented perforated plate screening, and widespread corrosion and rust staining are present on the perforated plate (See photo 3.1-50). The roof vent is not clog-resistant which is an AWWA and operational deficiency (AWWA D100-11 Sec. 7.5).

From the limited area observed during the ROV evaluation, the coating on the roof interior and support structure appears to be in fair overall condition as corrosion is located on areas of the roof and support structure (See photos 3.1-51 to 3.1-54). The roof interior support structure consists of radial roof rafters and a center column. The outer ends of the rafters were not visible during the ROV evaluation. The inner ends of the rafters are attached to a center hub which is located on the top of a column. The center column appears to be constructed of two channels welded together. Spots of corrosion are located on the column (See photo 3.1-52). The column base was not visible during the ROV evaluation. Lugs are located on the center column (See photo 3.1-53). Note: the lugs should not be used for rigging purposes.

From the limited area observed during the ROV evaluation, the coating on the interior shell appears to be in fair to poor condition as corrosion and pitting which had occurred prior to the previous repainting are located on areas of the shell (See photos 3.1-56 and 3.1-57). Dirt is present on the shell.

Float equipment attached to pipe brackets extending down from the roof is located on the interior of the tank (See photo 3.1-55). It was not determined if the float equipment is operating properly during the ROV evaluation.

The overflow inlet was not visible during the ROV evaluation. Cursory calculations indicate that approximately 8 in. (203 mm) would be required to provide adequate freeboard. As the overflow inlet was not visible during the ROV evaluation, it could not be determined if the existing freeboard satisfies the seismic requirements.

The interior floor surfaces were not visible during the ROV evaluation.

PHOTO 3.1-1 Tank 558 and site.

PHOTO 3.1-2 Tank 558 and site.

PHOTO 3.1-3 Building adjacent to the tank.

PHOTO 3.1-4 Fenced building and antenna tower on site.

PHOTO 3.1-5 Valve vault access.

PHOTO 3.1-6 Valve vault piping covered with jacketed insulation and corrosion on uncovered piping.

PHOTO 3.1-7 Valve vault piping covered with jacketed insulation, corrosion on uncovered piping, and water on the vault floor.

PHOTO 3.1-8 Partially buried foundation, and corrosion on bottom plates and stiffener.

PHOTO 3.1-9 Minor corrosion and metal loss along bottom plates and stiffener. Note grass clippings and leaves on the bottom plates.

PHOTO 3.1-10 Capped pipe projecting from the shell between the two bottom plates.

PHOTO 3.1-11 Pipe covered with jacketed insulation extending from building and penetrating the tank shell.

PHOTO 3.1-12 Pipe equipped with a valve and covered with jacketed insulation penetrating the tank shell. Note unplugged weep hole in the reinforcing plate.

PHOTO 3.1-13 Pipe covered with jacketed insulation and wiring extending from the valve vault.

PHOTO 3.1-14 Wiring extending from the valve vault connected to a threaded and plugged coupling in the tank shell.

PHOTO 3.1-15 Pipe equipped with a valve and covered with jacketed insulation penetrating near the bottom of the shell, shell manhole, and pipe bollards.

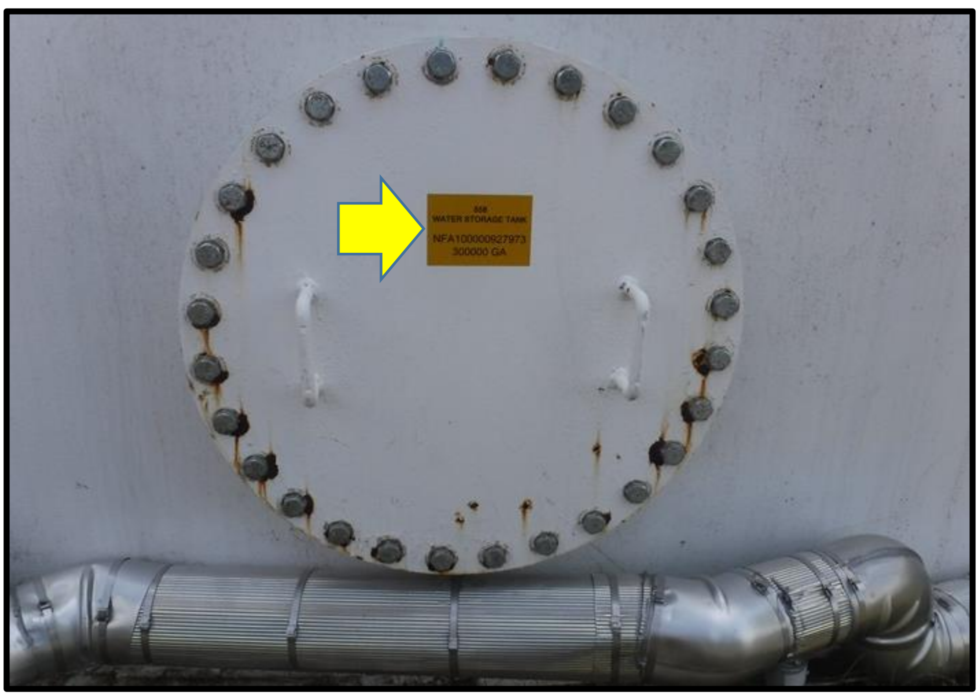


PHOTO 3.1-16 Shell manhole. Note tank information decal on the shell manhole cover.

PHOTO 3.1-17 Tank information decal on the shell manhole cover.

PHOTO 3.1-18 Welded steel insert plate in the shell around the shell manhole and pipe penetrating near the bottom of the shell.

PHOTO 3.1-19 Threaded and plugged coupling in the shell.

PHOTO 3.1-20 Unused studs and coating failure, corrosion, and rust staining on area of previous cabinet on the shell exterior.

PHOTO 3.1-21 Unused stud remains and coating failure, corrosion, and rust staining on area of previous cabinet bracket on the shell exterior.

PHOTO 3.1-22 Unused stud remains and coating failure, corrosion, and rust staining on areas of previous cabinet brackets on the shell exterior.

PHOTO 3.1-23 Conduit and brackets on the shell exterior.

PHOTO 3.1-24 Conduit extending up the shell and brackets on the shell exterior.

PHOTO 3.1-25 Conduits extending up the shell adjacent to the exterior shell ladder.

PHOTO 3.1-26 Corrosion on conduit bracket.

PHOTO 3.1-27 Stub overflow pipe equipped with perforated plate screening.

PHOTO 3.1-28 Corrosion and rust staining on the exterior shell ladder.

PHOTO 3.1-29 Corrosion on exterior shell ladder rungs and brackets.

PHOTO 3.1-30 Corrosion and rust staining on the exterior shell ladder.

PHOTO 3.1-31 Exterior shell ladder and safety cage.

PHOTO 3.1-32 Roof access and safety railing.

PHOTO 3.1-33 Electrical cabinet attached to the roof safety railing and water level indicating device within uncovered enclosure on the roof.

PHOTO 3.1-34 Capped water level indicating device equipment equipped with electrical outlets within uncovered enclosure on the roof.

PHOTO 3.1-35 Wiring penetrating circular hand hole cover plate on the roof.

PHOTO 3.1-36 Corrosion on the interior of the roof manhole cover and curb.

PHOTO 3.1-37 Corrosion on the roof manhole curb and wiring penetrating the curb.

PHOTO 3.1-38 Wiring extending from conduit junction box penetrating circular hand hole cover plate on the roof.

PHOTO 3.1-39 Conduit extending across the roof exterior.

PHOTO 3.1-40 Wiring extending from conduit junction box penetrating circular hand hole cover plate on the roof.

PHOTO 3.1-41 Corrosion on areas of the roof exterior.

PHOTO 3.1-42 Corrosion on areas of the roof exterior.

PHOTO 3.1-43 Corrosion on areas of the roof exterior. Note threaded and plugged coupling and bolted rectangular plate on the roof exterior.

PHOTO 3.1-44 Sealant around bolt in the roof.

PHOTO 3.1-45 Sealant around rusty stud in the roof.

PHOTO 3.1-46 Corrosion on bolted rectangular plate on the roof exterior. Note deteriorated gasket.

PHOTO 3.1-47 Corrosion on threaded and plugged coupling on the roof.

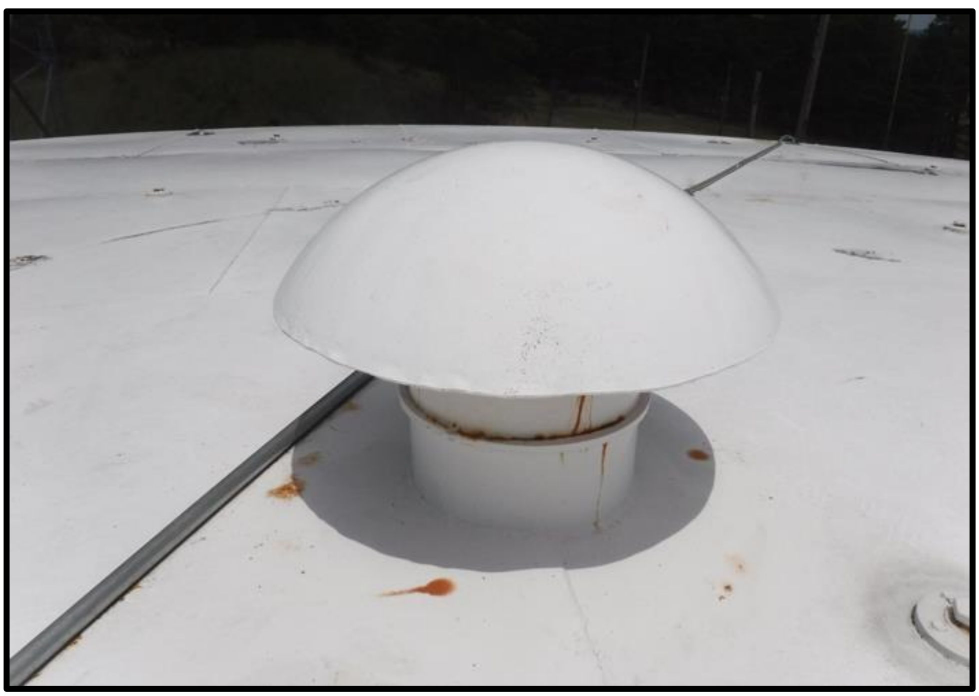


PHOTO 3.1-48 Roof vent.

PHOTO 3.1-49 Rust staining on the roof vent neck and corrosion along stiffener.

PHOTO 3.1-50 Widespread corrosion on roof vent perforated plated screening.

PHOTO 3.1-51 Roof interior and support structure.

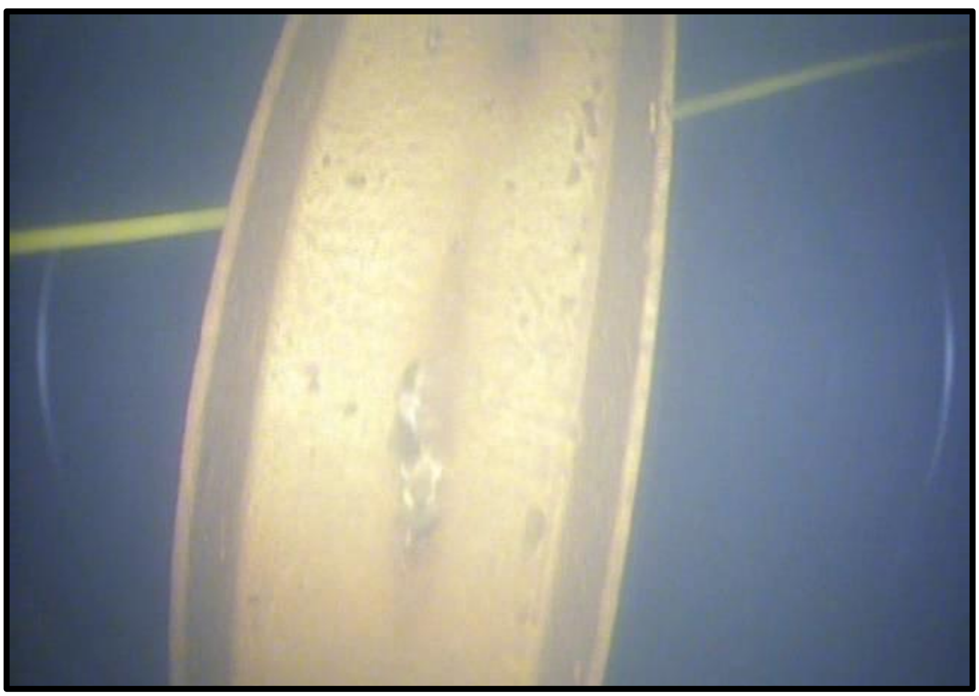


PHOTO 3.1-52 Spots of corrosion on the center column.

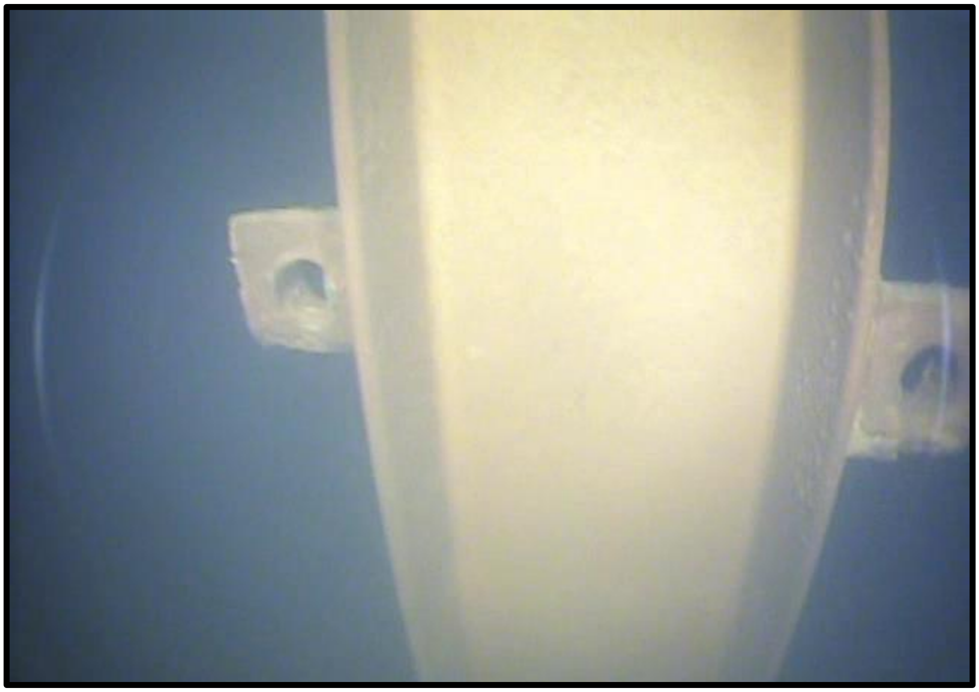


PHOTO 3.1-53 Lugs on the center column.

PHOTO 3.1-54 Roof interior and support structure.

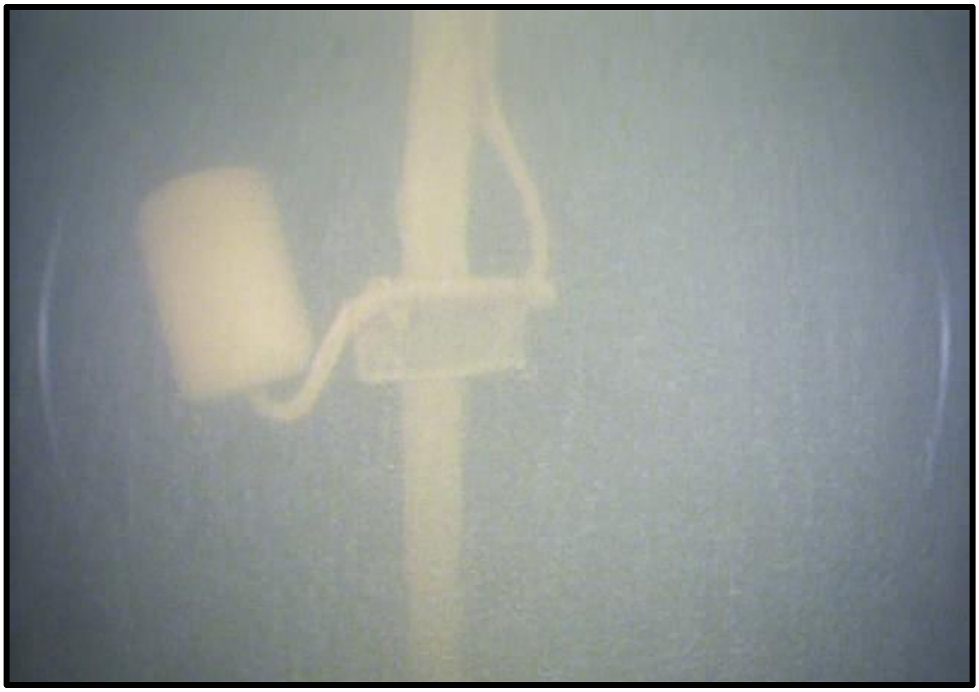


PHOTO 3.1-55 Float equipment inside the tank.

PHOTO 3.1-56 Pitting in shell interior.

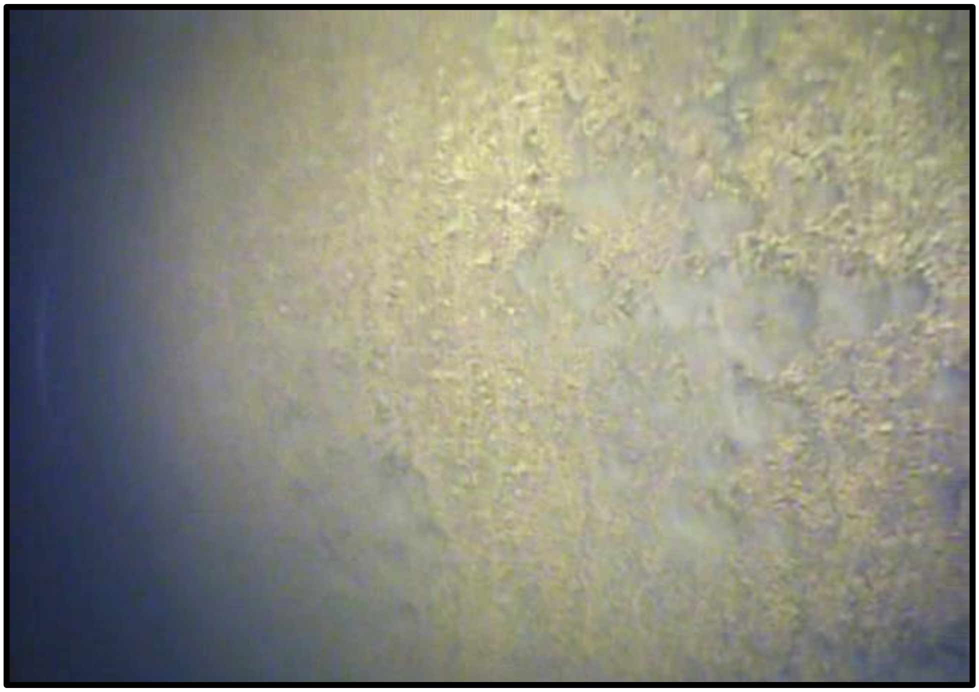


PHOTO 3.1-57 Dirt on the shell interior.

3.1.3 Comparison of Previous Inspection Results

No previous evaluation reports are provided.

3.1.4 Structural Condition Assessment

The conditions reported herein reflect the condition of the tank as observed on the date of the evaluation, using reasonable care in making the observations, and safety in gaining access to the tank. Should latent defects be discovered during the cleaning of the structure, they should be brought to the attention of the Navy and the Engineer.

This tank is located in a region of moderate seismic activity. This evaluation and the reporting of the condition of this tank do not warrant the structural condition of the tank or any of the original design for seismic loadings. Likewise, recommendations for this tank do not include modifications which may be required for compliance with present structural codes. It is possible the tank was erected in compliance with pre-existing industry standards which have since been replaced by more restrictive standards.

3.1.5 Recommendations

Foundation and Site:

Site Maintenance: The Navy should consider installing a chain link fence topped with barbed wire around the immediate tank site for improved tank and site security. The site should be regraded so that the top of the foundation projects a minimum of 6 in. (152 mm) above grade and so that proper drainage away from the foundation continues. Site maintenance should be performed with the mower discharge directed away from the base of the tank to prevent rock chips in the coating and the accumulation of grass and leaves on the bottom plates.

Foundation: When the next exterior recoating is performed, any unsound concrete should be chipped to sound material and the concrete should be brush-off blasted. Any deteriorated areas or voids found should have a bonding agent and a vinyl emollient modified concrete patching mortar applied to build up the surface to its original contour. The concrete should then be painted with a concrete sealer.

Install Sealant: When the exterior repainting is performed, a flexible polyurethane sealant should be installed between the foundation and lower bottom plate.

Valve Vault: A sump with pump should be installed inside the valve vault to eliminate the water on the vault floor. The jacketed insulation on the piping should be removed, and the piping inspected to determine its condition. The piping and valves located in the valve vault should be cleaned and painted in accordance with the interior coating recommendations at the time of the tank cleaning and coating. The exterior concrete surfaces should be cleaned to the equivalent of a brush-off blast cleaning and painted with a concrete sealer. The valve vault access should be locked at all times in order to limit liability to the Owner and to protect water system security. Freeze protection should be provided on all control piping and static water lines.

Exterior Surfaces:

Life of the Exterior Coating: The coating on the exterior of the tank appears to be in fair to poor overall condition as corrosion is located on several areas of the roof and shell. Tank Industry Consultants recommends the exterior should be recoated within the next 2 to 3 years.

Coating Testing: Prior to preparation of specifications for the cleaning and coating of the exterior of the tank, samples of the exterior coating system should be subjected to laboratory analysis to test for ingredients which may at that time be subject to regulations concerning their handling and disposal.

Cleaning: When the exterior is to be cleaned, all varieties of containment should be investigated. Containment of the wind-blown debris spent cleaning and paint overspray may will be required due to the proximity of the adjacent buildings.

Recommended Coating System:

Complete Cleaning and Repainting: The optimum long-life coating system presently available for this site is an zinc-polyurethane coating system, AWWA D102-17 Outside Coating System No. 6 (OCS-6). Properly formulated and applied polyurethanes have good resistance to condensation, mildew, and chipping. The polyurethanes also have excellent color and gloss retention and the longest expected service life of any of the common exterior tank coatings. The use of zinc-primed coatings may increase the cost of exterior tank recoating. However, this type of coating system normally eliminates the requirement of completely removing the coating system for future exterior tank recoating. The typical life of a properly applied epoxy-polyurethane coating system is approximately 15 to 20 years. These coatings also presently meet current VOC requirements.

Coating Application: The entire shell exterior should be cleaned to the equivalent of an SSPC-SP 6, Commercial Blast Cleaning and have an epoxy-primed, epoxy intermediate and polyurethane finish coating system applied. Due to the presence of mill scale, an SSPC-SP 10, Near-White Blast Cleaning will likely be required. However, care must be taken during the application of this particular coating system because this coating does have poor dry-fall characteristics, and potential damage to the surrounding property must be taken into consideration. The polyurethane coatings also require close monitoring of temperature and humidity during application.

Effective Service Life: Tank Industry Consultants defines the life of a coating as the amount of time before repainting becomes necessary due to coating failure and corrosion. During the coating life the Navy should expect the coating to lose its gloss, start to chalk, show signs of weathering, and possibly some rust staining. Future touch-up may be required on isolated coating failures. If aesthetics are a concern, the Navy may have to topcoat the repainted tank prior to the end of the expected service life. However, future topcoating would be less expensive than complete cleaning and recoating and could delay the next complete cleaning and repainting for many years.

Other Systems: With air emission volatile organic compounds (VOC) restrictions being put in place around the nation, alternative coating systems may become available which would be viable options for this tank. The Navy should review the available systems prior to preparing specifications for the recoating project.

Coating Curing: It would be more economical to paint the tank exterior at the same time the interior is painted, since the tank must be drained while the exterior is painted, and the applied coatings cure. This will also reduce mobilization and observation costs.

Rehabilitation Schedule: To obtain the lowest possible prices for the work outlined in the recommendations, the Navy should have the specifications prepared and the work bid in the spring, with the work scheduled to start in early summer, if possible.

Grinding and Bracket Removal: Any unused brackets, studs, or erection lugs should be removed prior to the exterior repainting. Any weld burrs, weld spatter, or erection scars should be ground off to provide a smooth surface for the application of the coating. This includes the unused stud remains located on the shell.

Electrical Apparatus: All unused electrical conduit, fixtures, and electrical metering equipment should be removed from the tank and tank site. All required equipment should be repaired and maintained in accordance with the National Electric Code (NEC).

Level Indicating Device: The electrical outlets at the level indicating device equipment on the roof should be equipped with ground fault interrupt circuits. The proper operation of the level indicating device should be verified. If the Owner does not wish to operate or replace the level indicating device floats and wires, all associated couplings, brackets, and components should be removed from the tank. Patch plates should be welded over the openings created by the indicating device removal. The patch plates should be completely seal welded on both the interior and exterior surfaces.

Exterior Piping: The jacketed insulation on the exterior piping penetrating the tank should be removed, and the piping inspected to determine its condition. The unplugged weep hole in the reinforcing plate around the pipe penetration on the west side of the shell should be tapped and plugged.

Existing Shell Manhole: At the time of recoating and repairs, the gasket for the shell manhole should be replaced, and the bolts and nuts equipped with new galvanized washers. The existing shell manhole should be equipped with a hinged support arm located on the shell exterior.

Additional Shell Manhole: Tank Industry Consultants interprets OSHA standards as defining a water storage tank as a confined space, and as such, a second means of emergency egress and ventilation during cleaning and coating or inspection operations is required to achieve compliance with OSHA, safety-related standards, and TIC's recommended best practices. Tank Industry Consultants recommends that tanks have two shell manholes or similar entry/egress points (at least one of which is a minimum of 30 in. in diameter) so "Acceptable Entry Conditions" as defined in 29 CFR 1910.146 can be maintained during tank entry for evaluation, repair, cleaning, and other interior tank service activities. The availability of two entry/egresses points also provides the Owner means to meet their obligations for the safety requirements stated in 29 CFR 1910.146 (c)(8). The presence of a second shell manhole would provide an unobstructed means of entry/egresses during interior tank service activities during which hoses, cords, fans and other standard equipment are present in one of the manhole openings. AWWA Standard D100 also requires a second shell manhole. Therefore, the tank should be equipped with a second hinged shell manhole. The additional manhole and cover should be 30 in. in diameter, should be designed in accordance with current industry and safety standards, should be equipped with a hinged arm located on the shell exterior, and should be located approximately 180 degrees from the existing shell manhole.

Overflow Pipe: Stub overflow pipes result in the potential for ice formation on the exterior of a tank as well as erosion of the soil due to effluent discharge. Such icing conditions can cause damage and accelerated corrosion and metal loss rates. Therefore, Tank Industry Consultants and the 10 State Standards recommend extending the pipe to near grade. The overflow pipe should exit the top shell ring and extend to approximately 24 in. above grade attached to the shell by welded steel brackets. The overflow pipe discharge could be equipped with a screened, counter-weighted flap gate or elastomeric check valve if desired by the Navy. The air break should be adequately sized to allow the proper functioning of the new flap gate. The overflow effluent should be directed away from the foundation using a concrete splash block and rip rap.

Exterior Ladder: A safety cage is not required on ladders with safe-climbing devices. To reduce cleaning and painting costs and future maintenance costs, and because OSHA is phasing out the use of safety cages in lieu of the installation of safe-climbing devices, the cage could be removed and a safe-climbing device installed. The exterior ladder did not include slip-resistant rungs. Slip-resistant rungs are required for all ladders constructed after March 1991 by the OSHA Construction standards and is considered to be best practice. However, slip-resistant rungs are not required by the OSHA General Industry standards for ladders or by AWWA D100.

Vandal Deterrent: When the existing safety cage is removed, a new vandal deterrent should be installed. Installing a vandal deterrent would offer the Navy further protection from unauthorized access to the ladder and tank.

Roof Safety Railing: The roof safety railing access opening should be equipped with a self-closing gate as is now required by OSHA. The roof safety railing toe bar should be lowered such that the gap between the toe bar and the roof is less than 1/4 in. (6 mm) as required by OSHA.

Cover Plates on Roof: The missing or rusty bolts on the bolted cover plates on the roof should be replaced. The existing gaskets at the cover plates should be replaced with new ones.

Clog-Resistant Vent: The tank is not equipped with a clog-resistant vent. AWWA Standards recommend that all vents with screening against insects be designed to ensure "fail-safe" operation if the insect screens become occluded. Inadequate ventilation could cause a tank collapse if the tank is rapidly drained while the screen is occluded or frosted over. Therefore, a clog-resistant vent should be installed near the center of the roof. The existing vent opening should be equipped with a large flanged opening for evaluating, cleaning, repairing, and painting the interior roof structure ends and the center hub. The flanged opening will then be used to mount a clog-resistant vent at the center of the roof. The vent should be designed so that it is removable in order to access the ends of the roof rafter and center hub.

Existing Roof Manhole: The roof manhole and cover should be locked to improve water system security.

Additional Roof Manhole: Best practices recommend a second roof manhole for emergency egress during coating and repairing operations. Therefore, if the existing vent is not replaced with one attached to a 24 in. (610 mm) diameter flanged opening, a second roof manhole should be installed in the roof. The manhole and cover should be designed in accordance with current industry and safety standards. The new roof manhole should be installed between roof support structure members to allow unrestricted use of the manhole. Both the new and the existing roof manholes should be locked at all times to prevent unauthorized access to the tank interior.

Roof Hand Holes: If a hanging anode cathodic protection system will not be installed and used in the future, then the openings should be covered with welded steel patch plates when repainting the tank.

Interior Surfaces:

Life of the Interior Coating: The interior surfaces should be drained and inspected before the rehabilitation of the tank. The interior coating appears to be in fair to poor overall condition as corrosion is located on areas of the roof, support structure, and shell. Tank Industry Consultants recommends the interior should be recoated within the next 2 to 3 years. When the interior is repainted, an epoxy coating system should be used.

Coating Testing: Prior to preparation of specifications for the cleaning and coating of the interior of the tank, samples of the interior coating system should be subjected to laboratory analysis to test for ingredients which may at that time be subject to regulations concerning their handling and disposal.

Recommended Interior Coating System:

Coating System: The standard Navy water tank interior coating system is a three-coat ANSI/NSF Standard 61 approved, polyamide epoxy system, designed in accordance with NFGS 09972.

Coating Application: The standard Navy surface preparation for interior surfaces, as described in NFGS 09972, is SSPC-SP 5, White Metal Blast Cleaning. NFGS 09972 also requires dehumidification during all surface preparation, coating application, and coating initial cure.

Service Life: The typical life of a properly formulated and applied epoxy coating system is approximately 12 to 15 years in immersion service. Tank Industry Consultants defines the life of a coating as the expected service life before repainting becomes necessary due to coating failure and corrosion.

Cathodic Protection: When the tank is rehabilitated the brackets and fittings should be installed for the future installation of a cathodic protection system.

Type: When the cathodic protection system is installed, an ice-resistant cathodic protection system that features long-life anodes, automatic potential and current control should be specified.

Scheduling: After the interior is completely cleaned and recoated, the cathodic protection system should not be energized until after the First Anniversary Evaluation. The Navy should conduct washouts and evaluations approximately every 3 years to monitor the need for cathodic protection. As the interior coating begins to show signs of failure, the cathodic protection system should be energized to aid in minimizing corrosion below the top capacity level.

Maintenance: Cathodic protection, if used and maintained properly, will control active corrosion below the water level and extend the useful life of a coating system. It should be noted that maintenance as recommended by the cathodic protection manufacturer is required for the cathodic protection system to work properly. Without proper monitoring, the cathodic protection system may operate too high and cause the coating to blister, or the system may operate too low and not adequately protect the exposed steel surfaces.

Rough Edges: All unused brackets should be removed from the interior and exterior surfaces at the time of the next recoating. Any weld burrs, spatter, scars or rough edges in the steel should be ground smooth to provide a better surface for coating.

Pit Welding and Pit Filling: After initial cleaning, all significant pitting should be welded, and all pitting with rough edges that would make the pitting difficult to coat properly should be filled with a solventless epoxy seam sealer.

Seam Sealing: The existing roof manhole and existing roof vent intersections should be sealed with an epoxy seam sealer at the time of the interior recoating.

Roof Support Structure: After initial abrasive blast cleaning, the roof support structure should be carefully evaluated as metal loss repairs may be necessary at areas where the metal loss was not previously visible.

Overflow Inlet: The interior of the tank should be drained and inspected to determine if the existing overflow inlet satisfies the current seismic requirements. Cursory calculations indicate that approximately 8 in. (203 mm) freeboard is required. This could be achieved by lowering maximum operating level or adding shell plates/raising roof. The addition of shell height may necessitate additional foundation and anchorage modifications and could be cost prohibitive.

3.2 CLOSURE:

The recommended work should be performed by a competent bonded contractor, chosen from competitive bids taken on complete and concise specifications. The coatings used should be furnished by an experienced water tank coating manufacturer to supply the field service required for application of technical coatings.

All work performed and coatings applied should be in accordance with NACE, ANSI/NSF Standard 61, the manufacturer's recommendation, ASCE 10, ACI, AWWA D100, D102, and (latest revisions), and SSPC: The Society for Protective Coatings.

Observation of the work in progress by experienced personnel will offer additional assurance of quality protective coating application. Observations can be performed on a continuous basis or be intermittent, capturing critical phases. The actual cost of observation may be less using intermittent as opposed to continuous observation; however, with intermittent observation it is often necessary for work to be redone to comply with the specifications. This negatively impacts the quality of the finished product, lengthens the job, and is frequently a cause of conflict between the contractor, client, and field technician. Continuous observation minimizes the amount of “rework” performed which improves schedule and overall business relationships.

An anniversary evaluation should be conducted immediately prior to the end of a one year bonded guarantee. Washouts and coating, structural, sanitary, safety, and corrosion evaluations should be conducted not less than every three years.

If the recommended work is not performed within the next three years, the structures should be reevaluated prior to the preparation of specifications and solicitation of bids.

The recommendations in this report are not intended to be specifications on which a contractor can bid. Complete bidding documents must include general and special conditions, detailed technical specifications, and other information necessary for the competitive bidding process. To properly protect the interests of the client, contractor, and engineer; the initial evaluation, the technical specifications, legal portions of the contract documents, and the observation should be performed by the same firm or with close coordination of all parties involved.

APPENDIX A

KEY PERSONNEL

KEY PERSONNEL

Point of Contacts (POC):

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APPENDIX B

INSPECTION PROCEDURES

PREBID EVALUATION:

The first step of a tank maintenance program is an outlined engineering evaluation of the condition of the tank. The tank evaluation herein consists of a study of the tank's interior, exterior, foundation (visible), and accessories. It includes on-site field measurements and observations, as well as evaluations of the structural members of the tank, and coatings, safety, and sanitary conditions. The result is the submittal of a certified engineering report outlining our observations and recommendations for rehabilitation and maintenance.

FIELD EVALUATION:

During the field evaluation, technicians accessed the tank's interior and exterior surfaces looking for sanitary, safety, and structural deficiencies, which were then analyzed by professional tank engineers. Ultrasonic steel thickness measurements are taken to enable structural engineers to analyze deviations from original steel thicknesses and assess structural deficiencies if deemed necessary.

Various tests were performed and samples taken to provide information which was vital in determining recommendations for the painting and rehabilitation of the tank. These included:

- Measurements of the tank members
- Measurements of the tank accessories
- Measurements of metal loss
- Coating thickness
- Ultrasonic steel thickness
- Noting of irregularities or unusual circumstances
- Photographic documentation

COATING EVALUATION:

Coating samples were taken during the field evaluation and were tested to determine their lead, chromium, and cadmium content. Coating thickness and adhesion tests were performed during the coating evaluation to offer insight into the "topcoat-ability" of the existing coating.

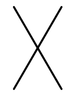
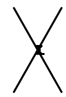
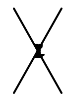
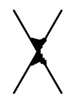
STRUCTURAL EVALUATION:

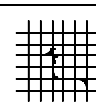
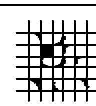
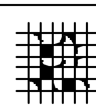
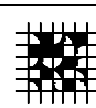
Included in this evaluation and report is the identification of observed structural deficiencies and/or damage which may have occurred since the tank was erected. These observed deficiencies could be deviations from the original design and construction and/or deterioration which may have occurred since the construction of the tank. This evaluation and the reporting of the condition do not warrant the original structural condition or any original design for seismic loadings. Likewise, recommendations for this tank does not include modifications which may be required for compliance with present structural codes.

ADHESION TESTS:

Adhesion tests were performed in accordance with ASTM D3359. The results are reported herein using the ASTM scale. The ASTM scale is a relative scale to rate adhesion from 0 to 5, with 5 being the best. A table of adhesion test results classifications is included in this Appendix.

Classification of Adhesion Test Results

Method A – X Cut Tape Test Approx. 1.5 in. long cuts at 30 deg. to 45 deg. apart.	Surface	Classification
No peeling or removal.		5
Trace peeling or removal along incisions.		4
Jagged removal along incisions up to 1/16 in. (1.6mm) on either side.		3
Jagged removal along most of incisions up to 1/8 in. (3.2mm) on either side.		2
Removal from most of the area of the X under the tape.		1
Removal beyond the area of the X.		0

Method B – Lattice Cut Tape Test Six parallel cuts at 2mm apart.	Surface	Classification
The edges of the cuts are completely smooth; none of the squares of the lattice are detached.	No Failure	5
Small flakes of the coating are detached at intersections; less than 5% of the lattice is affected.		4
Small flakes of the coating are detached along edges and at intersections of cuts. The area affected is 5% to 15% of the lattice.		3
The coating has flaked along the edges and on parts of the squares. The area affected is 15% to 35% of the lattice.		2
The coating has flaked along the edges of cuts in large ribbons and whole squares have detached. The area affected is 35% to 65% of the lattice.		1
Flaking and detachment worse than grade 1.		0

ASTM 3359 Standard Test Methods for Measuring Adhesion by Tape Test

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APPENDIX C

STRUCTURAL DATA

TANK 558

ULTRASONIC THICKNESS MEASUREMENTS:

(all readings were taken through coating)

Roof Plates:	0.273 in. to 0.276 in. (7 mm)
Shell:	
Ring #3:	0.289 in. to 0.301 in. (7.6 mm)
Ring #2:	0.283 in. to 0.287 in. (7.3 mm)
Ring #1:	0.319 in. to 0.329 in. (8.4 mm), bottom
Upper Bottom Plate:	0.346 in. to 0.361 in. (9.2 mm)
Lower Bottom Plate:	0.456 in. to 0.464 in. (11.8 mm)

FOUNDATION AND SITE:

TANK SITE:

Size: approx. 102 ft (31.1 m) x 117 ft (35.7 m) immediate area
Fence: none

Adjacent Structures:

Type: fenced antenna tower and building
Direction: west
Distance: approx. 4 ft 3 in. (1.3 m)

Type: Building 559
Direction: south
Distance: approx. 10 ft 3 in. (3.1 m)

Nearest Overhead Power Lines:

Direction: west
Distance: approx. 52 ft (15.8 m)

FOUNDATION:

Type: concrete
Projection Above Grade:

North: below grade
South: 0 in. to 4 in. (102 mm)
East: below grade
West: below grade

Grout: none
Sealant: none

VALVE VAULT:

Location: south of tank
Size: 4 ft (1.2 m) x 22 ft (6.7 m) x 4 ft 9 in. (1.4 m), deep

TANK 558

Access:
Size: 4 ft (1.2 m) x 6 ft (1.8 m)
Locked: no

EXTERIOR:

DESCRIPTION:
Construction: welded steel
Diameter: approx. 50 ft (15.2 m)
Shell Height: approx. 22 ft (6.7 m)
Number of Rings: 3
Roof Type: column and rafter supported

STENCIL:
Location: on shell manhole cover

558
Water Storage Tank
NFA 100000927973
300000 GA

BOTTOM PLATES:
Number: 2
Lower Bottom Plate Projection: 1 in. (25 mm) to 3 in. (76 mm) from shell
Upper Bottom Plate Projection: 2 in. (51 mm) to 3 in. (76 mm) from shell
Upper Bottom Plate Stiffener: 4 in. (102 mm) x 3/8 in. (10 mm), flat bar

ANCHOR BOLTS: none

SHELL MANHOLE:
Location: west side of shell ring #1
Type: flanged and bolted
Size: 24 in. (610 mm) diameter
Neck: 5 in. (127 mm) to 5-1/2 in. (140 mm) x 0.295 in. to 0.315 in. (8 mm) thick
Flange: 3-7/8 in. (98 mm) projection x 1/2 in. (13 mm) thick
Bolts:
Number: 28
Size: 3/4 in. (19 mm) square x 2-1/2 in. (64 mm) long
Cover:
Size: 32-3/4 in. (832 mm) diameter x 1/2 in. (13 mm) thick
Hinged: no

EXTERIOR PIPING:
Pipe 1:
Location: penetrates near the bottom of the shell on west side, extends to the south side of site, and extends into valve vault
Size: approx. 4 in. (102 mm) diameter

TANK 558

Pipe #2:
Location: penetrates south side of shell and extends into Building 559
Size: approx. 8 in. (203 mm) diameter

OVERFLOW PIPE:
Type: stub
Size: 4 in. (102 mm) diameter
Projection From Shell: 3 in. (76 mm)
Location: approx. 18 in. (457 mm) below the roof
Screening: perforated plate

EXTERIOR LADDER:
Number of Rungs: 21
Distance From Ground to Lowest Rung: 11 in. (279 mm)
Width: 16 in. (406 mm)
Side Rails: 1-1/2 in. (38 mm) x 2-1/2 in. (64 mm) x 1/8 in. (3 mm), thick
Rung Size: 1 in. (25 mm) diameter, smooth
Rung Spacing: 12 in. (305 mm) on center
Toe Room: 7-1/4 in. (184 mm)
Brackets:
Construction: bolted to ladder and welded to shell
Size: 2-1/4 in. (57 mm) x 1/8 in. (3 mm), flat bar x 3-1/2 in. (89 mm) long
Spacing: approx. 4 ft (1.2 m)
Safe-Climbing Device: none
Safety Cage:
Width: 27-1/4 in. (692 mm)
Depth: 27-1/4 in. (692 mm)
Vertical Bars:
Number: 7
Spacing: 8 in. (203 mm)
Size: 1 in. (25 mm) x 3/16 in. (5 mm), flat bar
Horizontal Bars:
Size: 1-1/4 in. (32 mm) x 3/16 in. (5 mm), flat bar
Spacing: 4 ft (1.2 m)
Vandal Deterrent: none

ROOF SAFETY RAILING:
Handrail:
Height: 41-3/4 in. (1060 mm)
Size: 2-1/2 in. (64 mm) x 2-1/2 in. (64 mm) x 1/4 in. (6 mm), angle
Uprights: 2-1/2 in. (64 mm) x 2-1/2 in. (64 mm) x 1/4 in. (6 mm), angle
Mid-Rail: 2-1/2 in. (64 mm) x 2-1/2 in. (64 mm) x 1/4 in. (6 mm), angle
Toe Bar:
Size: 4 in. (102 mm) x 1/4 in. (6 mm), flat bar
Gap Between Toe Bar and Roof: 1/2 in. (13 mm)

TANK 558

Access Opening:
    Width: 28-1/4 in. (718 mm)
    Self-Closing Gate: no

ROOF OPENINGS:
  Manhole:
    Type: hinged
    Size: 29-1/2 in. (749 mm) x 30-1/2 in. (775 mm)
    Neck: 4-3/4 in. (121 mm) x 3/16 in. (5 mm), thick
    Welded: exterior and interior
    Overlap: 2-1/4 in. (57 mm) x 3/16 in. (5 mm), thick
    Locked: no

  Roof Vent:
    Type: mushroom cover
    Neck Diameter: 12 in. (305 mm)
    Neck Height: 12 in. (305 mm) x 0.445 in. to 0.456 in. (11.6 mm) thick
    Screening:
      Orientation: horizontal
      Type: perforated plate
    Cover: approx. 26-1/2 in. (673 mm) diameter x 3/16 in. (5 mm) thick

TABLE C.1-1
Tank 558 Exterior Coating and Metal Condition

	Coating Thickness		Approx. % Failure to		Adhesion	Metal Loss	
	Range	Typical	Underlying Coating	Rust		Typical	Deepest
Shell	7.3 mils to 20.8 mils (186 μm to 530 μm)	17.2 mils (439 μm)	Neg.	< 1/2%	3 T	Neg.	Neg.
Roof	13.5 mils to 17.4 mils (344 μm to 444 μm)	14.8 mils (377 μm)	Neg.	1% to 3%	4 T	Neg.	Neg.

Key to Table

Adhesion 5 (very good) T = Topcoat to Underlying Coating Neg. = negligible
4 (good)
3 (fair) S = Primer to Steel
2 (poor)
1 (very poor)
0 (very poor)

TANK 558
EVALUATION OF STEEL GROUND WATER TANKS
TABLE C.1-2

FOUNDATION AND SITE-RELATED CONDITION RATING	
Site drainage 0- Deep depressions and water around foundations and/ or site saturated long after rain 1- Mild depressions around foundations and/or water ponds adjacent to tank 2- Water flows toward foundation 3- Water ponds on site, but not adjacent to foundation 4- Flat site 5- Ground well sloped away from foundations	4
Foundation Projection and Settlement 0- Concrete foundation tops buried 1- Concrete foundation tops even with ground and covered with vegetation and debris 2- Concrete foundation tops even with ground and clean 3- Concrete foundation tops with 1 in. to 2 in. projection 4- Concrete foundation tops with 2 in. to 6 in. or 12 in. to 24 in. projection 5- Concrete foundation tops with 6 in. to 12 in. projection	0
Concrete Condition 0- Concrete soft, crumbling and/or easily removed and/or hollow sound when tapped Exposed rebar that is corroded 1- Same as 0 except foundation still has basic original shape and/or Exposed rebar with minor surface corrosion and/or open cracks up 1/4" wide 2- Surface crumbling, but can be chipped to sound concrete, open cracks up to 1/8" wide 3- Some surface spalling, but solid sounding and/or deep cracks up to 1/16" wide Hollow sounding areas 4- Minor surface cracks or deficiencies 5- Smooth surface with no cracks or deteriorations	4
Grout Condition / Sealant 0- Grout greater than 40% gone 2- Grout 20 to 40% gone 3- Grout 5 to 20% gone 4- Some pieces can be removed and/or grout less than 5% gone 5- Cracks and minor chips, but intact	5

TANK 558
TABLE C.1-2 (Cont'd)

CHIME AND ANCHORAGE CONDITION RATING	
Base of Shell 0- Metal loss greater than 50% T Spot pitting of 50% to 75% T 1- Metal loss of 25% to 50% T Spot pitting of 25% to 50% T Rust over 50% to 75% of surface 2- Metal loss up to 25% T Spot pitting less than 25% T Rust over 25% to 50% of surface 3- Metal loss not exceeding 10% T Rust over 10% to 25% of surface 4- General surface rust with only slight metal loss or minor pitting 5- No visible deterioration	5
Chime Projection 0- Holes in steel, bent plate Metal loss >75% T Cracked welds at bottom lap welds Water leaking from shell, shell-to-bottom joint or from beneath bottom 1- Metal loss greater than 50% T Spot pitting exceeding 75% T. 2- Chime projection less than 1/2" or thickness less than 1/8" 3- Metal loss greater than 25% T Spot pitting less than 25% T Rust over 25% to 50% of surface Soil or debris covers all or parts of chime 4- Metal loss not exceeding 25% T Deteriorated edges of chime General surface rust with only slight metal loss or minor pitting. 5- No visible deterioration	4
Anchor Bolts 0- Missing nuts Metal loss >1/3 original diameter 1- Metal loss on nuts of 1/4 T or greater Metal loss on bolts of 1/4 to 1/3 original diameter 2- Metal loss on nut <1/4 T Metal loss on bolt of 1/8 to 1/4 original diameter 3- Loose nuts or scaled rust on bolts 4- Paint failure and minor surface rust on bolts 5- No visible deterioration No anchors	5
Anchor Bolts and Chairs 0- Missing or cracked welds Missing or corroded rivets >50% 1- Bent top plate 3- Corrosion and/or debris inside chairs. 5- Chairs clean and well painted No anchors	5

TANK 558
TABLE C.1-2 (Cont'd)

SHELL LADDER CONDITION RATING	
Steel Condition 0- Metal loss on rung, side rail, or bracket of over 40% T Ladder distorted or bent 3 in. or more from original position. 1- Metal loss of 25% to 40% T Missing attachment bolts or cracked welds General instability 2- Metal loss of 10% to 25% T Loose bolts 3- Metal loss of 5% to 10% T 4- Metal loss of less than 5% T, or surface rust 5- No deterioration or no shell ladder	4
Safety Device 0- Device improperly attached and/or in danger of coming loose No safety device present or device inoperable 2- Present safety device works rough 5- Safety device installed properly and in good condition No shell ladder	5
Dimensions 0- 13 in. or less clear distance between side rails Rungs greater than 12 in. center-to-center. Lacks 7 in. toe clearance or 30 in. head clearance 3- 13 in. to 15 in. clear distance between side rails Side rails less than 2 in. x 3/8 in. 4- Side rails less than 2-1/2 in. x 3/8 in. 5- 16 in. or greater clear distance between side rails No shell ladder	5
Ease of Climbing 0- Major obstructions limiting climbing surface 1- Difficult transition to balcony (i.e. must climb over handrail) 2- Improper or erratic rung spacing 3- Minor obstructions connected to side rails 5- Ladder clear of obstructions and easy to climb No shell ladder	3

TANK 558
TABLE C.1-2 (Cont’d)

SHELL CONDITION RATING	
Coating Condition 0 - Exposed steel area more than 40% 1 - Exposed steel area 20% to 40% 2 - Exposed steel area 10% to 20% 3 - Exposed steel area 5% to 10% Exposed underlying coating area more than 40% Poor coating appearance 4 - Exposed steel area 1% to 5% Exposed underlying coating area 5% to 40% 5 - Exposed steel area less than 1% Exposed underlying coating less than 5% area	4
Steel Condition 0 – Spot pitting over 50% T Rust over 75% or more of the surface Upper course steel thickness less than 1/8” General metal loss over 3” diameter and larger or 8” and longer Suspect repairs if shell plate exceeds 1/2” (square door corners, lap patches) Shell nozzles >4” diameter 1 – Spot pitting of 25% to 50% T Rust over 50% to 75 % of surface Weld spacing around nozzles <3” for standard tanks or <4” for high strength steel 2 – Spot pitting less than 25% Rust over 25% to 50% of surface 3 – Rust over 10% to 25% of surface 4 – Rust over 1% to 10% of surfaces 5 – Rust over less than 1% of surfaces	4
Shell Manhole 0- Manhole leaking water 1- Only one manhole less than 20 in. diameter 2- Only one manhole with dimensions not less than 20 in. diameter 3- Two or more shell manholes but none 30 in. diameter or larger Missing or parts 5- Two or more shell manholes (one 30 in. diameter and others not less than 20 in. diameter)	2
Overflow 0- More than 1/3 of the support brackets broken Pipe clogged or broken Holes of greater than 1 in. diameter in pipe 1- Less than 1/3 of the brackets broken Holes smaller than 1 in. diameter in pipe Overflow does not have a weir box or anti-vortex plate Stub overflow pipe 2- Ineffective protective screen No screen or flap gate on discharge end of pipe Pipe does not extended to grade Overflow discharges through perimeter roof vent No visible air break 3- No splash block below overflow discharge 4- Rust on overflow pipe 5- No rust or deterioration	1

TANK 558
TABLE C.1-2 (Cont'd)

ROOF CONDITION RATING	
Coating Condition 0 - Exposed steel area more than 40% 1 - Exposed steel area 20% to 40% 2 - Exposed steel area 10% to 20% 3 - Exposed steel area 5% to 10% Exposed underlying coating area more than 40% Poor coating appearance 4 - Exposed steel area 1% to 5% Exposed underlying coating area 5% to 40% 5 - Exposed steel area less than 1% Exposed underlying coating less than 5% area Aluminum dome roof or no roof	3
Steel Condition 0- Spot pitting over 50% T or holes in roof Rust over 75% or more of surface or leak Steel thickness less than 1/8" over 10"x10" area Significant roof distortions over large are 1- Spot pitting of 25% to 50%T Rust over 50% to 75% of surface 2- Spot pitting less than 25%T Rust over 25% to 50% of surface Minor roof distortion 3- Rust over 10% to 25% of surface 4- Rust over 1% to 10% of surface 5- Rust over less than 1% of surface No roof (open top tank)	3
Roof-to-Vent Connection 0- Finial or vent connection failed / disconnected Holes greater than 1" diameter Cathodic protection hand holes open or covers missing Evidence of water entry into tank 1- Holes around finial or vent less than 1" diameter 2- Metal loss around vent or finial of 50% T or greater Missing bolts or cracked welds 3- Rust and/or metal loss of less than 50% T 4- Minor corrosion at connection 5- No rust or deterioration at connection – or no roof (open top tank)	5
Roof Vent and Roof Opening 0- No vent present Vent clogged or inoperative Screen missing, torn, or improperly placed Not clog-resistant 1- Vent deteriorated or damaged 2- Vent covered with rust Bolts or parts missing or loose 4- Minor rust on vent 5- No rust or deterioration – or no roof (open top tank)	0
Roof Manway 0- Manway without locking features or lock missing 1- Allows water into tank from roof 3- Any dimension less than 24 in. diameter, 4 in. curb or 2 in. overlap 5- Manway locked and water tight All dimensions equal or greater than 24 in. diameter, 4 in. curb or 2 in. overlap No roof (open top tank)	0

TANK 558
TABLE C.1-2 (Cont'd)

ROOF SAFETY RAILING CONDITION RATING	
Steel Condition 0- Metal loss on rung, side rail, or bracket of over 40% T Ladder distorted or bent 3 in. or more from original position. 1- Metal loss of 25% to 40% T Missing attachment bolts or cracked welds General instability 2- Metal loss of 10% to 25% T Loose bolts 3- Metal loss of 5% to 10% T 4- Metal loss of less than 5% T, or surface rust 5- No deterioration No roof ladder	5
Safety Device 0- Device improperly attached and/or in danger of coming loose No safety device present or device inoperable No self-closing gate 2- Present safety device works rough 5- Safety device installed properly and in good condition No roof ladder	0
Dimensions 0- 13 in. or less clear distance between side rails Rungs greater than 12 in. center-to-center. Lacks 7 in. toe clearance or 30 in. head clearance Safety railing members do not satisfy OSHA loading requirements 3- 13 in. to 15 in. clear distance between side rails Side rails less than 2 in. x 3/8 in. 4- Side rails less than 2-1/2 in. x 3/8 in. Gap between toe bar and roof more than 1/4 in. 5- 16 in. or greater clear distance between side rails No roof ladder	4
Ease of Climbing 0- Major obstructions limiting climbing surface 1- Revolving ladder 2- Improper or erratic rung spacing 3- Minor obstructions connected to side rails 5- Ladder clear of obstructions and easy to climb No roof ladder	5

TANK 558
TABLE C.1-2 (Cont'd)

<table>
<tr><th colspan="2">INTERIOR BOTTOM CONDITION RATING</th></tr>
<tr><td>Coating Condition
0- Exposed steel over > 40% of the area
Blistering to steel over 50% of the area
1- Exposed steel over 20% to 40% of the area
Blistering to steel over 25% to 50% of the area
2- Exposed steel over 10% to 20% of the area
Blistering to steel over 20% to 25% of the area
Exposed primer over greater than 40%.
3- Exposed steel over 5% to 10% of the area
Blistering to steel over 10% to 20% of the area
Exposed primer over 25% to 40%
4- Exposed steel on less than 5% of the area
Blistering to steel on less than 10% of the area
Exposed primer over 10% to 25%.
5- Exposed primer on less than 10% of the surface</td><td>5</td></tr>
<tr><td>Steel Condition
0- Holes in steel or cracked welds
Evidence of leakage
Metal loss exceeding 3" diameter or 6" long, exceeding 1/8" deep
Rivet heads with 50% deterioration
Corroded shell-to-bottom fillet weld less than 1/4" in size
Edge settlement exceeding API-653 limits
General corrosion or pitting w/ remaining thickness of 1/8" or less
1- Isolated pitting over 1" diameter w/ remaining thickness of 1/8" or less.
Groove-type pitting exceeding 9" long w/ remaining thickness of 1/8" or less
Metal loss exceeding 3ft x 3ft area w/ remaining thickness of 3/16" or less.
General metal loss over the entire bottom exceeding 1/16"
Rivet heads with 25% deterioration
Edge settlement exceeding 50% of API-653 limits.
2- Widely scattered pitting (less than 3 per square foot) exceeding 1/8" deep
Scattered groove-type pitting exceeding 9" long of 1/16" to 1/8" deep.
3- Widely scattered pitting (less than 3 per square foot) of 1/16" to 1/8" deep
Scattered groove-type pitting not exceeding 1/16" deep
Edge settlement up to 50% of API-653 limits
4- Scattered pitting not exceeding 1/16" in depth
Edge settlement up to 25% of API-653 limits
5- Scattered pitting not exceeding 1/32" in depth
No visible edge settlement</td><td>5</td></tr>
</table>

TANK 558
TABLE C.1-2 (Cont'd)

INTERIOR SHELL CONDITION RATING	
Coating Condition 0- Exposed steel over > 40% of the area Blistering to steel over 50% of the area 1- Exposed steel over 20% to 40% of the area Blistering to steel over 25% to 50% of the area 2- Exposed steel over 10% to 20% of the area Blistering to steel over 20% to 25% of the area Exposed primer over greater than 40%. 3- Exposed steel over 5% to 10% of the area Blistering to steel over 10% to 20% of the area Exposed primer over 25% to 40% 4- Exposed steel on less than 5% of the area Blistering to steel on less than 10% of the area Exposed primer over 10% to 25% 5- Exposed primer on less than 10% of the surface	4
Steel Condition 0- Holes in steel Upper shell steel thickness < 1/8" General metal loss >3" diameter General metal loss >8" long and 10% T General metal loss >8" long and 5% T (high-strength steel) Rivet head 50% deteriorated Cracked or missing shell welds 1- Isolated pitting >1" diameter exceeding 50% T Groove type pitting > 8" long and 25% to 50% T Rivet head 25% deteriorated General metal loss of 25% T in any 3' x 3' area General metal loss of 10%T over large areas 2- Widely scattered pitting (< 3 per square foot) over 50% T Scattered groove-type pitting > 8" long and 25% to 50% T 3- Widely scattered pitting (<3 per square foot) of 25% to 50% T Scattered groove-type pitting < 25% T 4- Widely scattered pitting not exceeding 25% T 5- Widely scattered metal loss not exceeding 10% T	4

TANK 558
TABLE C.1-2 (Cont'd)

INTERIOR ROOF CONDITION RATING	
Coating Condition 0- Exposed steel over 40% Blistering to steel over 50% of the area 1- Exposed steel over 20% to 40% of the area Blistering to steel over 20% to 50% of the area 2- Exposed steel over 10% to 20% of the area Blistering to steel over 20% to 25% of the area Exposed primer over 40% or more 3- Exposed steel over 5% to 10% of the area Blistering to steel over 10% to 20% of the area Exposed primer over 25% to 40% 4- Exposed steel on less than 5% of the area Blistering to steel on less than 10% of the area Exposed primer over 10% to 25% 5- Exposed primer on less than 10% of the surface	4
Steel Condition 0- Spot pitting over 50% T or holes in roof Steel thickness less than 1/8" over a 10"x10" area Rivet heads 50% deteriorated Cracked welds Rafters missing or ineffective Visible distortion of girders or bowing of columns Bolts missing or severely corroded, welds cracked 1- Spot pitting over 25% T and over 1" diameter Groove-type pitting > 8" long and 25% to 50% T Rivet heads 25% deteriorated General metal loss of 25% T in any 3 ft x 3 ft area Column base plates welded to tank bottom 2- Widely scattered pitting (< 3 per square foot) over 50% T Scattered groove-type pitting > 8" long and 25% to 50% T Columns visibly out-of-plumb 3- Widely scattered pitting (< 3 per square foot) of 25% to 50% T Scattered groove-type pitting of < 25% T General metal loss of 10% T in the entire roof 4- Widely scattered pitting not exceeding 25% T 5- Scattered metal loss not exceeding 10% T	4

TANK 558
TABLE C.1-2 (Cont'd)

INTERIOR LADDER CONDITION RATING	
Steel Condition 0- Metal loss on rung, side rail, or bracket of over 40% T Ladder distorted or bent 3 in. or more from original position 1- Metal loss of 25% to 40% T Missing attachment bolts or cracked welds General instability 2- Metal loss of 10% to 25% T Loose bolts 3- Metal loss of 5% to 10% T 4- Metal loss of less than 5% T, or surface rust 5- No deterioration No ladder	5
Safety Device 0- Device improperly attached and/or in danger of coming loose No safety device present or device inoperable 2- Present safety device works rough 5- Safety device installed properly and in good condition No ladder	5
Dimensions 0- 13 in. or less clear distance between side rails Rungs greater than 12 in. center-to-center. Lacks 7 in. toe clearance or 30 in. head clearance 3- 13 in. to 15 in. clear distance between side rails Side rails less than 2 in. x 3/8 in. 4- Side rails less than 2-1/2 in. x 3/8 in. 5- 16 in. or greater clear distance between side rails No ladder	5
Ease of Climbing 0- Major obstructions limiting climbing surface 1- Difficult transition to balcony (i.e. must climb over handrail) 2- Improper or erratic rung spacing 3- Minor obstructions connected to side rails 5- Ladder clear of obstructions and easy to climb No ladder	5

TANK 558
TABLE C.1-2 (Cont'd)

PAINTING SCHEDULES CONDITION RATING	
Exterior Surface Schedule 1- 0 to 1 years 2- 2 to 4 years 3- 5 or more years	2
Interior Surface Schedule 1- 0 to 1 years 2- 2 to 4 years 3- 5 or more years	2

TANK 558 SUMMARY CONDITION EVALUATION STANDARDS
TABLE C.1-3

External Summary Steel Ground Water Tank

Foundations

Site drainage	4	
Foundation projection	0	Below Grade in Areas
Concrete condition	4	
Grout condition	5	

Chime and Anchorage Condition

Base of Shell	5
Chime Projection	4
Anchor bolts	5
Anchor bolt chairs	5

Shell Ladder

Steel condition	4
Safety device	5
Dimensions	5
Ease of climbing	3

Shell Condition

Coating condition	4
Steel condition	4
Container manhole	2
Overflow	1

Roof Condition

Coating condition	3	
Steel condition	3	
Roof-to-vent	5	
Roof vent/opening	0	Not Clog-Resistant
Roof manway	0	Not Locked

Roof Safety Railing

Steel condition	5	
Safety device	0	No Self-Closing Gate
Dimensions	4	
Ease of Climbing	5	

Internal Summary Steel Ground Water Tank

Interior Bottom

Coating Condition	5
Steel Condition	5

Interior Shell

Coating condition	4
Steel condition	4

Interior Roof

Coating condition	4
Steel condition	4

Interior Ladders

Structural condition	5
Safety device	5
Dimensions	5
Ease of climbing	5

Painting Schedules

Exterior	2
Interior	2

Condition Evaluation Ratings

Safety Rating	78
Sanitary Rating	N/A
Structural Rating	83
*Exterior PCM Rating	37
*Interior PCM Rating	58
Overall Rating	**64**

*PCM = Painting, Corrosion, & Maintenance

APPENDIX D

PERTINENT BACKGROUND INFORMATION

NOMENCLATURE:

The terms used in describing the various components of water tanks are unique to the industry. In fact, the terms vary from firm to firm and from person to person. In an attempt to define the terms used in this report, a sketch of the general type of the tank covered is included. **Warning: Some appurtenances on these tanks may be referred to as erection or rigging attachments, lugs, or brackets. <u>This does not mean that they are safe for weight loading, and personnel fall protection</u>. Each attachment for each tank should be evaluated on an individual basis by a structural engineer or an experienced rigger before being used. These devices may have been intended for <u>only the original erectors</u> and painters to use with specialized equipment**.

HEAVY METALS TESTS:

Samples of the coating systems were taken and sent to a laboratory for Inductively Coupled Plasma-Atomic Emission Spectrometry analyses. The test results are as follows:

TABLE D-1
Heavy Metal Test Results from Tank 558

	Cadmium		Chromium		Lead	
	mg/kg	percent	mg/kg	percent	mg/kg	percent
Exterior	ND	ND	4.6	0.00046%	190	0.019%
Interior	2.6	0.00026%	39	0.0039%	350	0.035%

Tank Industry Consultants performs these tests only to determine if there is lead, cadmium, or chromium present in the coating samples. To limit damage to the existing coating, only small areas were tested. The small number of samples taken and the difficulty of retrieving all primer from the steel profile may cause the tests performed to not accurately represent the total coating system. Variations in thickness, types of coatings applied, and the interim cleaning and painting operations will also affect the actual readings. The reliability of the results is also dependent on the amount of primer included in the sample. Additional testing to determine the amount of leachable contaminants present in the spent cleaning debris will need to be performed following cleaning operations at the time of repainting. Results from the laboratory analysis follow.

EMSL Analytical, Inc.
6340 CastlePlace Dr., Indianapolis, IN 46250
Phone: (317) 803-2997 Fax: (317) 803-3047 Email: indianapolislab@emsl.com

Attn: **Bruce Hobbs**
Tank Industry Consultants
7740 West New York Street
Indianapolis, IN 46214

Phone: (317) 271-3100
Fax: (317) 271-3300

7/2/2021

The following analytical report covers the analysis performed on samples submitted to EMSL Analytical, Inc. on 6/25/2021. The results are tabulated on the attached data pages for the following client designated project:

Tank 558

The reference number for these samples is EMSL Order #162113712. Please use this reference when calling about these samples. If you have any questions, please do not hesitate to contact me at (317) 803-2997.

Approved By:

Allison Ford

Allison Ford, Chemistry Lab Manager

Chemical Testing
2845.25

EMSL Analytical, Inc.
6340 CastlePlace Dr., Indianapolis, IN 46250
Phone/Fax: (317) 803-2997 / (317) 803-3047
http://www.EMSL.com indianapolislab@emsl.com

EMSL Order:	162113712
CustomerID:	TICO62
CustomerPO:	
ProjectID:	

Attn: **Bruce Hobbs**
Tank Industry Consultants
7740 West New York Street
Indianapolis, IN 46214

Phone:	(317) 271-3100
Fax:	(317) 271-3300
Received:	6/25/2021 10:40 AM
Collected:	6/21/2021

Project: **Tank 558**

Analytical Results

Client Sample Description 1 Ext — ***Collected:*** 6/21/2021 — ***Lab ID:*** *162113712-0001*

Method	*Parameter*	*Result*	*RL*	*Units*	*Prep Date & Analyst*		*Analysis Date & Analyst*	
METALS								
3050B/6010D	Cadmium	ND	0.54	mg/Kg	6/30/2021	WF	6/30/2021	WF
3050B/6010D	Chromium	4.6	1.4	mg/Kg	6/30/2021	WF	6/30/2021	WF
3050B/6010D	Lead	190	1.4	mg/Kg	6/30/2021	WF	6/30/2021	WF

Client Sample Description 2 Int — ***Collected:*** 6/21/2021 — ***Lab ID:*** *162113712-0002*

Method	*Parameter*	*Result*	*RL*	*Units*	*Prep Date & Analyst*		*Analysis Date & Analyst*	
METALS								
3050B/6010D	Cadmium	2.6	0.76	mg/Kg	6/30/2021	WF	6/30/2021	WF
3050B/6010D	Chromium	39	1.9	mg/Kg	6/30/2021	WF	6/30/2021	WF
3050B/6010D	Lead	350	1.9	mg/Kg	6/30/2021	WF	6/30/2021	WF

Definitions:

MDL - method detection limit
J - Result was below the reporting limit, but at or above the MDL
ND - indicates that the analyte was not detected at the reporting limit
RL - Reporting Limit (Analytical)
D - Dilution Sample required a dilution which was used to calculate final results

APPENDIX G

REFERENCES

REFERENCES

1. AWWA D100 Welded Steel Tanks For Water Storage
2. AWWA D102 Coating Steel Water Storage Tanks
3. AWWA D103 Factory-Coated Bolted Carbon Steel Tanks for Water Storage
4. OSHA Regulations (Standards – 29 CFR)
5. MIL-HDBK 1110/1
6. ANSI/NSF Standard 61
7. SSPC (Society for Protective Coatings)
8. NACE (National Association of Corrosion Engineers)
9. UFC 3-310-04 Change 1 (January 27, 2010)
10. ASCE 7-16: Standard Minimum Design Loads for Buildings and Other Structures